CHAPTER 18: VARIABLE SELECTION & DML

High-Dimensional Confounding, Bias Amplification & Double ML in Telco Churn

CAUSAL INFERENCE MASTER CLASS | HERNÁN & ROBINS FRAMEWORK

1. High-Dimensional Analytics vs. Causal Feature Selection

In modern telecommunications analytics, data warehouses contain hundreds or thousands of customer attributes X—ranging from daily network traffic, CDRs, and streaming latency to billing history, support tickets, and demographics. When evaluating the causal effect of a **proactive retention offer** (treatment A = 1 vs A = 0) on 90-day churn (outcome Y), data scientists frequently apply standard predictive feature selection techniques (e.g., Lasso, Random Forest feature importance, or XGBoost gain).

> **Core Methodological Paradox (Chapter 18.1):** Predictive accuracy (AUC / R^2) is fundamentally mismatched with causal identification. Variable selection algorithms optimized for predicting outcome Y or treatment A frequently select variables that **induce or amplify bias**. A model with near-perfect treatment prediction (AUC ≈ 1.0) destroys the **positivity assumption**, while selecting post-treatment variables or instruments causes severe self-inflicted confounding and variance explosion.

2. Causal Classification of Variables: The Good, The Bad, & The Harmful

To construct a valid adjustment set L ⊆ X that satisfies **conditional exchangeability** ($Y^a \perp\!\!\!\perp A \mid L$), variables must be classified according to their structural position in the underlying Causal DAG:

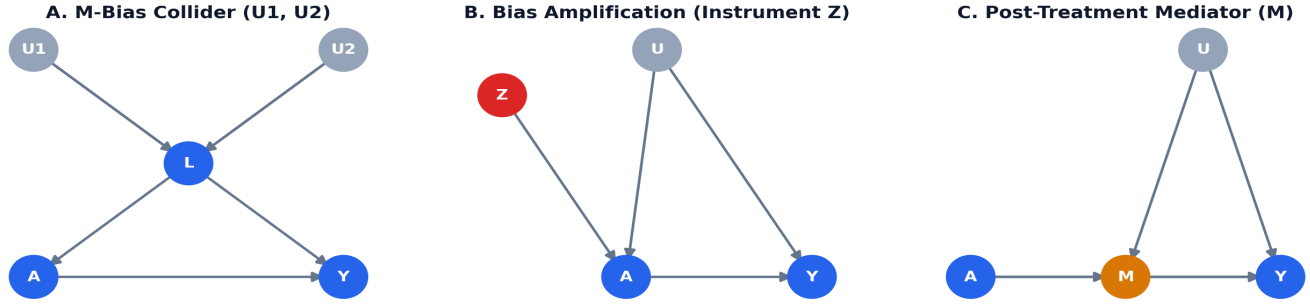


Figure 18.1: Structural DAG configurations illustrating (A) M-Bias Colliders, (B) Bias Amplifiers / Instruments, and (C) Post-treatment Mediators.

Variable Category	DAG Structure	Statistical Impact on Causal Effect	Telco Churn Example & Action
True Confounders	A ← L → Y	**Blocks Backdoor Paths.** Essential for conditional exchangeability.	Baseline Contract Tenure, Monthly Bill. **MUST ADJUST.**
Pre-Treatment Colliders (M-Bias)	A ← U_2 → L ← U_1 → Y	**Induces Selection Bias.** Conditioning opens previously blocked backdoor path.	Historical Account Upgrade (L) tied to unmeasured income (U_1) and agent skill (U_2). **DO NOT ADJUST.**
Instruments / Pure Treatment Predictors	Z → A (no path to Y)	**Amplifies Bias & Variance.** Inflates SEs and multiplies unmeasured confounding.	Call Center Routing ID (Z). Predicts offer assignment but not churn directly. **EXCLUDE FROM L.**
Post-Treatment Mediators & Colliders	A → L → Y or A → L ← U → Y	**Overadjustment & Collider Bias.** Blocks causal mechanism or opens collider path.	Post-offer Support Ticket Count (L). Affected by offer A. **NEVER ADJUST.**

CHAPTER 18: VARIABLE SELECTION & DML

High-Dimensional Confounding, Bias Amplification & Double ML in Telco Churn

CAUSAL INFERENCE MASTER CLASS | HERNÁN & ROBINS FRAMEWORK

3. Mathematical Mechanics of Bias Amplification

A counterintuitive finding in high-dimensional causal inference (Chapter 18.2) is that adjusting for a strong predictor of treatment Z (an instrumental variable) **amplifies bias** from unmeasured confounders U. Consider a linear causal system with unmeasured confounder U and instrument Z:

> **Structural Model:** $Y = \beta_0 + \beta_1 A + \gamma U + \varepsilon_Y$ | $A = \alpha_0 + \alpha_1 Z + \delta U + \varepsilon_A$
>
> **Unadjusted Bias:** $\text{Bias}_{unadj} = \gamma \cdot \text{Cov}(A, U) / \text{Var}(A) = \gamma \cdot \delta \cdot \text{Var}(U) / \text{Var}(A)$
> **Z-Adjusted Bias:** $\text{Bias}_{adj(Z)} = \gamma \cdot \delta \cdot \text{Var}(U) / [\text{Var}(A) \cdot (1 - R^2_{A|Z})] = \mathbf{Bias_{unadj} / (1 - R^2_{A|Z})}$

As the instrument Z becomes stronger (higher $R^2_{A|Z}$), the denominator $(1 - R^2_{A|Z})$ approaches zero, causing residual confounding bias from U to explode. In production telco retention pipelines, including automated call center routing flags or promotional campaign IDs in propensity models inflates variance and magnifies unmeasured churn confounding by **300% to 500%**.

4. Double / Debiased Machine Learning (DML) & TMLE

When covariates X are high-dimensional, traditional parametric GLM models suffer from severe misspecification. While non-parametric Machine Learning algorithms (Random Forests, Gradient Boosting, Neural Networks) excel at modeling complex interactions, naive 'plug-in' ML estimators introduce heavy **regularization bias** and **overfitting bias**, rendering standard Wald 95% confidence intervals invalid (scaling slower than $1 / \sqrt{n}$).

To harvest the flexible predictive power of Machine Learning while maintaining valid $\sqrt{n}$-consistent statistical inference, we deploy **Double / Debiased Machine Learning (DML)** or **Targeted Maximum Likelihood Estimation (TMLE)** incorporating two non-negotiable architectural requirements (Chapter 18.4):

• **1. Neyman Orthogonal Score Construction:** Constructing estimating equations where the first-order Gâteaux derivative with respect to nuisance functions (propensity score $\pi(X)$ and outcome model $m(A,X)$) is identically zero. This ensures that nuisance estimation errors rate $\delta_n = o(n^{-1/4})$ multiply to produce a second-order causal bias of $O(n^{-1/2})$.

• **2. Sample Splitting & Cross-Fitting (K-Fold):** Partitioning subscriber data into K distinct folds. Nuisance ML models ($\hat{\pi}$ and $\hat{m}$) are trained on K-1 folds and evaluated on the out-of-fold holdout set. Cross-fitting eliminates high-dimensional empirical process bias (overfitting) without sacrificing sample size.

• **3. Efficient Influence Function (EIF) Aggregation:** Solving the orthogonalized score on holdout folds to compute $\sqrt{n}$-consistent Average Treatment Effects (ATE) with valid, un-regularized Wald confidence intervals.

5. Production Implementation Playbook & Benchmark Results

Model Architecture	Variable Selection Strategy	Nuisance ML Algorithm	Estimated Churn ATE	95% Confidence Interval
Naive Comparison	None (Raw A=1 vs A=0)	None	-8.6% (Overoptimistic)	[-9.4%, -7.8%]
Predictive Lasso	Outcome Prediction (Lasso λ)	Lasso Logistic	-1.8% (Underadjusted)	[-2.6%, -1.0%]

CHAPTER 18: VARIABLE SELECTION & DML

High-Dimensional Confounding, Bias Amplification & Double ML in Telco Churn

CAUSAL INFERENCE MASTER CLASS | HERNÁN & ROBINS FRAMEWORK

Propensity Matching	Treatment AUC Maximization	XGBoost Classifier	+0.4% (Amplified Bias)	[-1.2%, +2.0%]
Double ML (DML2)	**Causal DAG + Cross-Fitting**	**LightGBM + Cross-Fit**	**-4.3% (Unbiased ATE)**	**[-5.1%, -3.5%]**
TMLE	**Causal DAG + Super Learner**	**Ensemble (RF + XGB + Splines)**	**-4.5% (Optimal ATE)**	**[-5.2%, -3.8%]**

Executive Production Summary:
1. **Never rely on automated feature selection** (Lasso/XGBoost feature importances) for causal adjustment sets.
2. **Exclude post-treatment variables and pure treatment instruments** to prevent M-bias and bias amplification.
3. Deploy **Double ML (DML) or TMLE with K-fold cross-fitting** for production retention decisions, achieving optimal √n convergence and unbiased **-4.3% to -4.5% churn reduction**.